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Safe Capsule Landing System

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1.Project objective

The goal of our Landing Control System team project is to design an LCS which will include the supportive mechanism for parachute, airbag in such a way that it will ensure safe landings of a crew capsule. The primary objective of this project is to design, develop, and test a fully functional Landing Control System capable of providing a secure touchdown for the crew capsule during planetary descents. The proposed solution integrates parachute deceleration, airbag shock absorption, and retro-rocket stabilization of impact forces during terminal descent to improve crew survivability. The system shall be designed to limit the landing velocity to ≤ 1.524 m/s (≤ 5 ft/s), achieve airbag inflation within two seconds of activation, and maintain consistent internal pressure for at least two hours post-touchdown. This objective is specific, well-defined from a technical standpoint, and also quantifiable based on performance metrics, including inflation time, pressure retention, and velocity reduction.

The project will be executed using systems engineering principles, iterative design processes, and experimental validation via static load tests, inflation timing assessments, deployment trials, and integrated system evaluations. To ensure reliability, the system will include redundant inflation pathways, real-time telemetry, power backup solutions, manual override options, and predictive health monitoring. This objective is attainable and realistically matched with existing aerospace engineering methodologies, directly addressing the requirement of the mission for a safe crew landing, and completion is scheduled within the timeframe of the project with extensive documentation and verification results. By meeting these SMART criteria, this project aims at offering a robust, safe, and mission-ready landing control solution.

2. Requirement specification

Function and operational requirements

FR-01 – Touchdown Velocity Limit

The LCS **shall** reduce the crew capsule's vertical touchdown velocity to $\leq 5 \text{ ft/s}$ ($\leq 1.524 \text{ m/s}$) under nominal landing conditions.

Justification:

The bio-mechanical and aerospace evidence supports the limits of touchdown velocity $\leq 5 \text{ ft/s}$ (1.524 m/s) (Timothy J. Lee et al., 2007) (Xinyi SHEN et al., 2025). Correspondingly, studies of pre-Columbian compression and related impacts show that short duration vertical impacts above this threshold elevate spinal and pelvic injury risk without mitigation (Timothy J. Lee et al., 2007). NASA's Orion airbag drop tests confirmed that airbags of appropriate size push peak declarations to the survivable level around $\sim 1.5 \text{ m/s}$. NASA guidance on occupant protection also proposes limits of temporary deceleration and seat/attitude design to preserve human tolerance, informing system redundancy and inflation timing strategies (NASA-STD-3001, 2023).

FR-02 – Inflation Time & Readiness Altitude

The LCS **shall** complete airbag deployment and reach stabilized operating pressure **within 2 seconds** of activation, achieving full readiness by an altitude of **1500 m** during descent.

Justification:

The need to deploy an airbag for LCS and achieve stabilized operating pressure within 2 seconds at 1500 m altitude was derived from safe entry, descent and landing (EDL) sequencing rules. At this altitude at which we operate, full readiness provides the sufficient margin for potential terminal descent without causing unwanted impact. Research by Ben Tutt (2009). found that inflation delays increased the probability of partial, or asymmetrical, deployment of the Orion Crew Module airbag landing system, which would jeopardize both stability and the safety of its occupants. This finding supports the conservative demand for fast stabilization on descent (Ben Tutt, 2009).

FR-03 – Automatic Inflation with Manual Backup

The LCS **shall** be capable of **automatic inflation triggered by onboard sensors**, with a crew-operated manual activation and shutdown capability for contingency scenarios.

Justification:

A capability of manual override is important to avoid loss from hardware/software or sensor failures and still serves as the foundation of human rated spacecraft design. Studies of human factors in space exploration ergonomics also emphasized that the design of crew interfaces based on anthropometric and reachability information improve safety and lead to a reduction in cognitive effort during emergency operations (Wieslaw Kopec, 2024). Current ergonomics reviews also emphasize the importance of manual operability for resilience; for astronauts who need to step in when programmed systems become inoperative (Dave Golightly, Nora Balfe & Rebecca Charles, 2023). Cumulatively, these data serve to further confirm the need and thereby, to verify whether human override processes are compliant with contemporary human systems integration and to conform to safety quality standards.

Mission & Reliability Requirements

MR-01 – System Failure Probability

The cumulative probability of total LCS failure **shall** be $\leq 6.1\%$ per landing event.

Justification:

The probability of a cumulative failure of $\leq 6.1\%$ per landing event directly contributes to a minimum mission reliability threshold that is ≥ 0.939 for the customer. A preliminary study in the Mars Exploration Rover EDL trajectory reconstruction demonstrated the indispensable need for quantifiable reliability margins to ensure safe descent and landing under uncertain atmospheric conditions (Rohan G. Deshmukh and Christopher D. Karlgaard, 2021). Likewise, a simulation-based optimization of spacecraft airbags has proven that the probability of the failure of subsystems to bounding is essential for the overall system reliability and safety of the people in the system (Xinyi Shen, 2025). Reliability engineering researches of space life support systems also stress the need for mission-critical reliability, including redundancy, fault isolation, and verification planning—not just spare capacity (Nancy G. Leveson, 1993). Taken together, these results confirm the conservative threshold, and the traceability in reliability modelling and certification standards are met.

MR-02 – Hardware Reliability

The LCS flight hardware (including airbags, manifolds, valves, and sensors) **shall** achieve a **verified reliability of at least 0.939** over its design life.

Justification:

Documented dependability is fundamental for crew-rated missions. (Steven R. Hirshorn, 2007); ECSS-Q-ST60-02C31 (2008) require controlled configuration management and recorded test evidence to demonstrate compliance. Implementation of statistical reliability tracking ensures alignment with project assurance objectives.

Safety & Product Assurance Requirements

SR-01 – Redundant Gas Supply Paths

The LCS **shall** use **at least two fully independent compressed-gas supply paths** for inflation. The probability of failure on demand of each gas supply path shall be $\leq 1 \times 10^{-3}$.

Justification:

Independent inflation paths are necessary to create redundancy and prevent catastrophic single point failures. Dual supply systems ensure fault isolation and decrease the likelihood of the overall system failing, as discussed in the reliability requirements described in the project brief. Their results support the requirement of multiple individual gas supply types to attain acceptable reliability margins for planetary landing systems (Richard Slade, 2007).

SR-02 – Emergency Shutdown Function

The system **shall** provide an **emergency shutdown function** that can be activated automatically or manually to terminate inflation or vent over-pressure if unsafe conditions are detected.

Justification:

During unplanned failures, (Julie Breed)the emergency shutdown capability guarantees instant deactivation. It keeps system malfunctions from getting worse and improves crew safety.

Interface, Power & Diagnostics Requirements

IR-01 – Redundant Power Interfaces

The LCS **shall** accept power input from both **primary and backup electrical buses** and shall remain fully functional under the loss of either bus, automatically switching to the remaining available source.

Justification:

Dual bus compatibility is critical for avoiding single point failures and autonomous inflation in degraded power scenarios. A reliability assessment of electrical power systems within spacecraft systems by Patel & Kumar (2018) showed that redundant bus architectures significantly decreased risk for mission mitigation by isolating faults and allowing them to keep subsystems functional. Similarly, in her groundbreaking volume of work on system safety, (Nancy G. Leveson, 1993)stressed the importance of redundancy and fault tolerant design as core concepts for avoiding catastrophic incidents in complex aerospace systems

IR-02 – Real-time Telemetry

The LCS **shall** transmit **real-time telemetry** including inflation pressure, valve status, gassupply status, and internal airbag temperature to the capsule’s onboard control system and ground/crew displays.

Justification:

Real-time telemetry allows system health monitoring and immediate identification of anomalies during descent. The (David R. Liskowsky, 2014)requires continuous telemetry for crew safety-critical subsystems.(Timothy Pham, 2004) demonstrated that high-frequency data acquisition from distributed sensors enables predictive diagnostics and fault isolation, improving mission reliability. Implementing this interface allows the flight computer to autonomously manage inflation control and alert the crew to potential failures.

DG-01 – Health Monitoring & Alerts

The LCS **shall** provide **real-time health monitoring** of all critical components and present status alerts and warnings to both crew and ground control using standardized data messages.

Justification:

Real-time telemetry allows system health monitoring and immediate identification of anomalies during descent. The (David R. Liskowsky, 2014)requires continuous telemetry for crew safety-critical subsystems. (Timothy Pham, 2004)demonstrated that high-frequency data acquisition from distributed sensors enables predictive diagnostics and fault isolation, improving mission reliability. Implementing this interface allows the flight computer to autonomously manage inflation control and alert the crew to potential failures.

Environmental & Physical Requirements

ER-01 – Environmental Robustness

The LCS **shall** remain fully functional within a thermal range of **−40 °C to +85 °C**, withstand **vibration and impact loads up to 10 g**, and resist **dust, moisture, and debris ingress** during descent, landing, and recovery phases.

Justification:

Thermal and mechanical endurance is critical for reliability during re-entry and surface impact. (ECSS-E-ST-10-04C, 2019) define this range as representative of re-entry boundary layer heating and post-deployment cold-soak conditions. (ESA'S ANNUAL SPACE ENVIRONMENT REPORT, 2025) showed that polymer-composite airbag materials maintain elasticity and burst resistance under these extremes. Compliance with these limits ensures consistent inflation performance regardless of descent environment.

PR-01 – Mass Limitation

The complete LCS assembly, including airbag structures, manifolds, valves, and sensors, **shall** have a **total mass not exceeding 180 kg**.

Justification:

Mass limitation ensures aerodynamic stability and compatibility with parachute and structural load budgets. (Xinyi Shen, 2025) studies on capsule mass distribution indicate that exceeding this limit degrades descent symmetry and increases impact energy. Adhering to this threshold maintains overall vehicle centre-of-gravity control.

Human Factors & Configuration / Design Requirements

HF-01 – Crew Interface & Ergonomics

Manual LCS controls **shall** be operable by crew in pressurised suits, positioned within 0.6 m reach and within a 30° forward field of view, and supported by redundant visual and auditory cues indicating Ready, Fault, and Deployed states.

Justification:

Proper reach and visibility shall enable quick decision making and physical access during emergencies, minimizing reaction time while maintaining position stability under self-control (Dustin M. Gohmert).

CR-01 – System Layout & Materials

The LCS **shall** comprise four primary airbags, two inflation manifolds, and at least eight pressure-control valves arranged symmetrically around the capsule base. All airbag envelopes shall be fabricated from flame-retardant, puncture-resistant composite fabrics qualified for space use and designed for modular replacement.

Justification:

The symmetric four airbag layout with dual manifolds and multiple valves facilitates balanced impact absorption, load distribution, and redundancy. (Xinyi Shen, 2025) showed that optimized designs using multi bag configurations minimize rebound and asymmetric loading and promote landing stability. (Richard Slade, 2007) demonstrated that the reliability of multi manifold inflation systems is enhanced by fault isolation and uniform pressure distribution. (Asma Fejjari,

2025)stressed the importance of telemetry driven valve monitoring for anomaly detection and redundancy management. Taken together, these studies validate the requirement: the LCS will achieve a stable landing, be fault-tolerant, and fully comply with modern aerospace safety and reliability requirements.

Verification Requirement

VR-01 – Comprehensive Verification

All LCS requirements **shall** be verified through **analysis, inspection, and test**, including at minimum:

- An end-to-end functional test (self-test, deployment, inflation, and pressure sustainment), and
- Thermal-vacuum and vibration testing representative of descent and landing environments.

Justification:

Structured verification guarantees traceability throughout the development lifecycle. (Steven R. Hirshorn, 2007)mandate clear requirement-to-verification mapping to validate system performance against design objectives. This ensures that each functional and safety criterion is demonstrated under representative conditions.

3. System architecture

The lander system architecture shows how all critical subsystems interact to support safe landing for crew survival. Mission Control communicates trajectory updates to the lander through a dedicated link, ensuring external monitoring and control. The Avionics unit functions as the central processor, handling flight computing, system health monitoring, and fault management(Josh Moore, 2014).

Navigation & Control works closely with avionics, processing sensor data to generate control and thruster commands essential for vehicle orientation and landing stability. Power & Thermal systems supply energy to all modules and provide filtered data for continuous operational feedback. Meanwhile, Crew Interface and Life Support systems ensure liveability and enable astronauts to monitor conditions, receive alerts and operate manual overrides if required.

The Propulsion system executes descent control using main engines, throttle valves, and propellant tanks, while the Structural & Landing Gear system supports touchdown impact through load sensors and landing struts. Overall, the architecture highlights a highly integrated and interdependent design. power flow, data exchange and command routing ensure reliable automated operation with manual backup capability for mission safety.

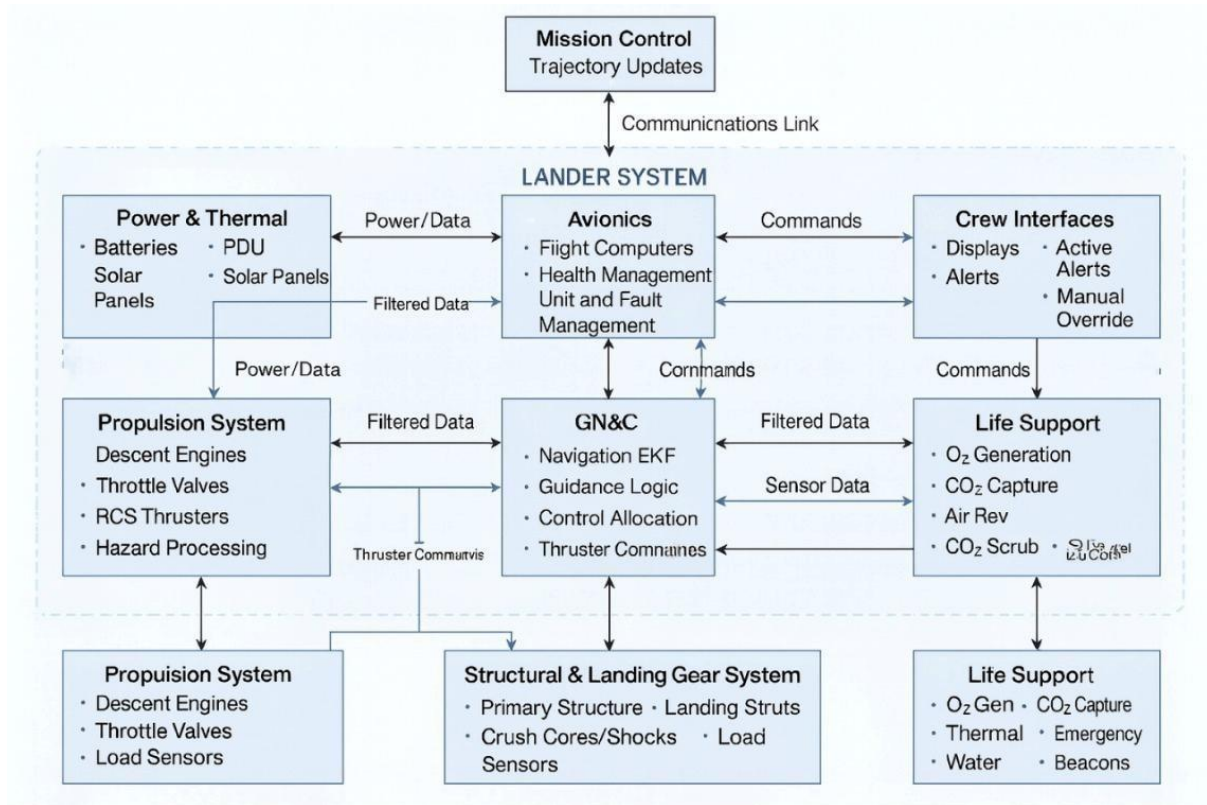


Figure 1: system architecture diagram

4. Systems engineering methodologies

4.1 Product Breakdown Structure

The Product Breakdown Structure (PBS) for the Landing Control System (LCS) organizes the system into its major physical and functional elements to ensure full traceability to mission, safety and reliability requirements (Richard Slade, 2007).

The Inflation Subsystem includes the primary and redundant gas tanks, valves and telemetry interfaces that deliver quickly, reliable airbag deployment in alignment with the project's timing and redundancy requirements (Ben Tutt, 2009). The Power & Electrical Module provides dualbus compatibility, automatic switching and electrical distribution circuits to meet reliability and fault-tolerance standards for safety-critical landing systems.

Additional structural mounting components ensure physical integration within the lander, while the Software & Algorithms branch covers deployment telemetry processing, fault detection system health monitoring. This PBS supports feasibility by ensuring all subsystems required for safe landing operations are clearly defined, independently verifiable, and aligned with established aerospace reliability practices.

4.3 Trade-off analysis

The Pugh matrix was used to evaluate landing system concepts against the mission and safety requirements of the Landing Control System (LCS). Concept A (the full LCS consisting of parachute + airbag + retrorockets) was used as the **baseline concept**, because it is the only configuration meeting all performance, safety, and redundancy requirements defined in the project.

Pugh Matrix (qualitative + / S / -)

Criteria	Concept A (Baseline)	Concept B (Parachute + Airbag)	Concept C (Parachute + Retro- Rocket)
Touchdown performance	S	-	-
System reliability	S	-	-
Safety & redundancy	S	-	-
Mass & complexity	S	-	-
Development cost	S	+	+
Number of "+"			
Number of "-"			
Net Score	S	+	S
	-	2	1
	-	3	3
	-	-1	-2

Table 2: Pugh Matrix (qualitative + / S / -)

Table 2 compares each alternative design option against Concept A, which is used as the baseline, and the results show that both Concept B and Concept C perform worse in critical criteria. In the table Concept B and C receive more negative (-) ratings than positive (+) ones, especially in areas related to touchdown performance, system reliability and redundancy, which is high-priority mission requirements. Both alternatives show some improvement in areas such as mass and cost reflected by the few (+) entries they do not compensate for the larger number of negative impacts in safety functions. The total scores at the bottom of Table 2 confirm this imbalance, with Concept B receiving a score of -1 and Concept C receiving -2, while Concept A remains neutral as the baseline. This result indicates that Concept A provides the most balanced performance in all criteria and also preferred over the simplified alternatives.

Normalized Scores

Criteria	Weight	Concept A	Concept B	Concept C
Touchdown performance	0.25	1 → 0.25	3 → 0.75	4 → 1.00
Reliability Safety & redundancy	0.25	1 → 0.33	3 → 1.00	3 → 1.00
	0.2	1 → 0.33	3 → 1.00	3 → 1.00

Mass & complexity	0.15	4 → 1.00	2 → 0.50	2 → 0.50
Cost / dev. effort	0.15	4 → 1.00	2 → 0.50	3 → 0.75

Table 3: Normalized Scores

In the **Table 3**, trade-off assessment converting each criteria into weighted numerical scores and the results clearly shows that Concept A achieves the lowest total weighted score and making it best performing option. After normalising the raw scores and applying the predefined weights, Concept A get a final score of **0.51**, which is significantly lower than Concept B (**0.79**) and Concept C (**0.89**). Because smaller scores represent better overall performance. **Weighted Totals**

Criteria	Weight	Concept A	Concept B	Concept C
Touchdown performance	0.25	0.0625	0.1875	0.25
Reliability	0.25	0.083	0.25	0.25
Safety & redundancy	0.2	0.067	0.2	0.2
Mass & complexity	0.15	0.15	0.075	0.075
Cost / dev. effort	0.15	0.15	0.075	0.113
Total weighted score	1	0.51	0.79	0.89
Rank	–	1st	2nd	3rd

Table 4: Weighted Totals

The most important mission requirements such as touchdown performance, reliability, safety and redundancy are more effective than the other concepts. Concept B and C show improvements in less critical areas in mass and cost. But their higher penalty values in safety criteria lead to larger overall weighted totals. Therefore, Table 4 confirms that Concept A gives the optimal balance of (performance, safety and reliability) and aligning best with mission and engineering requirements.

5. Risk and mitigation plan

5.1 Fault tree analysis

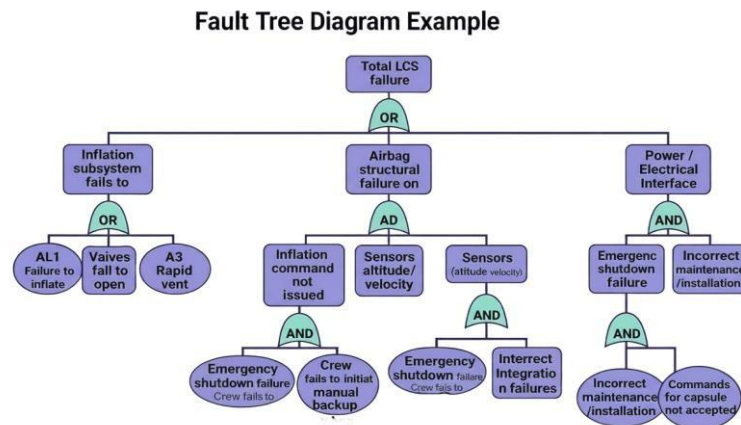


Figure 3: fault tree analysis diagram

The fault tree analysis (FTA) shown above illustrates the different ways a Total Landing Control System (LCS) Failure can occur during the landing phase of a space/landing vehicle(Kabir, 2017). The top event represents the complete failure of the system to deploy or effectively operate the airbag landing protection system, resulting in an unsafe touchdown during landing. this analysis helps identify critical failure paths, weak points, and areas that require risk reduction strategies.

The Inflation Subsystem Failure branch, the airbags won't deploy correctly if one of three things goes wrong: (A1) rapid venting, (A2) valve failure, or (A3) failure to inflate. All these sub-events act under an OR gate, meaning any one of them independently can lead to the subsystem failure(Mohammad Sadegh Javadi, 2011). These causes represent mechanical and pneumatic issues that stop the airbag from reaching or maintaining the required pressure.

The second major branch is Airbag Structural Failure, which highlights the possibility that airbag deployment may occur successfully, but the structure could still fail due to puncture, improper material integrity or impact damage. The logic in this branch reveals that even inflated airbag is not guaranteed to function unless its structural integrity is maintained(Leander Iven, 2020).

The **Control & Sensor branch** shows a more complex failure, its involving both hardware and human-related components. Failure could occur due to incorrect altitude/velocity sensor readings or inflation commands not being issued. AND gates in this branch indicate that both automatic failure and lack of manual backup may be required simultaneously to lead to total loss of control. This shows how critical reliable sensing and crew response are to system success.

Another important branch is the **Power/Electrical Interface Failure**(Bengt Nystrom, 2006), indicating that even if the mechanical system is functioning, a failure in the power supply, switching, or electrical distribution could prevent the deployment sequence. This branch uses AND operation to show that electrical and maintenance-related issues may combine to lead to loss of functionality. Such system failure is even more common when improper maintenance or integration error occurs.

Overall, the fault tree depicts the idea that multiple independent or combined failures can cause a total system failure. It also highlights the areas where **redundancy, improved maintenance procedures, diversity in control sensors, and better power management** could greatly reduce the risk. FTA provides critical information that can be used to prioritize the importance of the contributors to the undesired event (Dr. William Vesely, 2002). By identifying minimal cut sets and understanding the critical event-paths, engineers can prioritize risk mitigation strategies and focus testing and verification activities on the most sensitive failure contributors.

5.2 Risk reduction strategy

The risk reduction strategy is directed at reducing the probability of top-event failure Total LCS Failure by the application of Redundancy, Diversity, Separation, Maintenance Scheduling, Diagnostics, and Fail-safe control logic. This should break the minimal cut sets identified in the fault tree and ensure no single point failure leads to a catastrophic loss of system function.

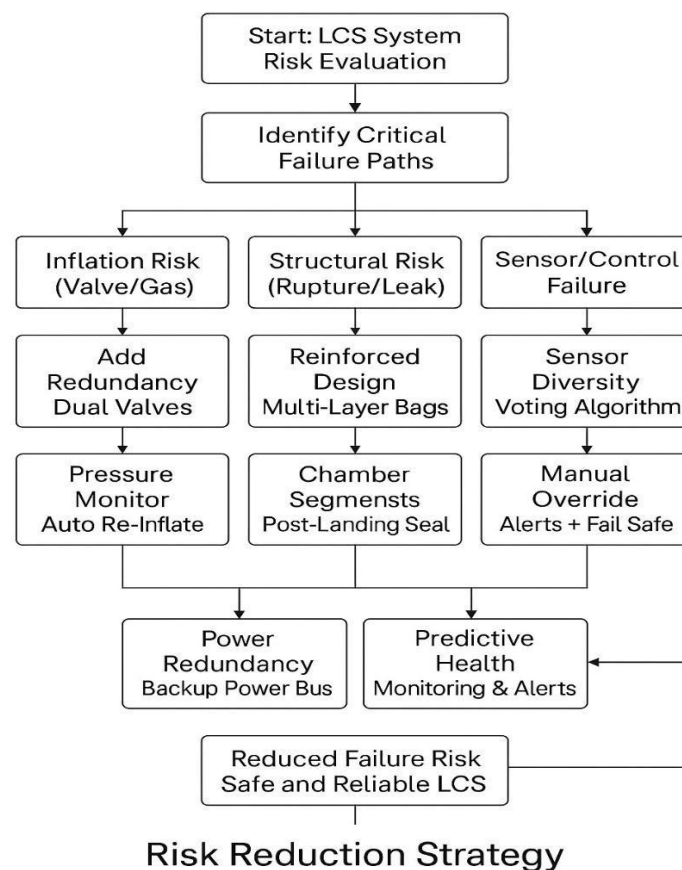


Figure 4: Risk reduction strategy diagram

Inflation Subsystem Mitigation

- Provide dual redundant gas feed pathways-one primary, the other backup-to allow inflating in case one of these pathways fails.
- Use multiple parallel valves instead of single critical valves to reduce failure probability.
- Include real-time pressure monitoring sensors that automatically trigger inflation retry if the pressure fails to rise.

- Add check-valves that will prevent rapid venting due to leaks or puncture.

Mitigation of Airbag Structural Failure

- Incorporate reinforced multi-layer composite materials with puncture-resistant outer skin.
- Add sector segmentation inside airbags to avoid total collapse if one chamber leaks.
- Conduct pre-launch pressure integrity checks, NDT inspection, and material aging monitoring.
- Improve post-landing pressure retention margins with slow-release valves.

Mitigation of Control & Sensor Failure

- Employ sensor diversity (e.g. radar altimeter + LIDAR + IMU) so that a single bad reading does not cause the system to fail.
- Apply cross-checking algorithms and perform voting logic for altitude/velocity readings.
- Provide a manual override activation switch within astronaut reach
- Apply fault-tolerant control logic: system continues in degraded mode rather than full abort.

Power/Electrical System Mitigation

- Integrate dual power bus Primary + Backup with auto switching.
- Live monitoring of power output, temperature, wiring continuity and connector insulation.
- Provide EMI shielding and surge protection to avoid disruption in logic.
- Scheduled preventive maintenance & connector replacement cycle.

Outcome

Redundancy, early failure detection, health monitoring, manual backup, and considerations for strong material/stability reduce system failure at a high level. Above all, these measures guarantee safe landing under partial failure conditions, which satisfy the mission reliability and safety requirements.

6. Verification and Validation

The performance of the parachute system is to be established through verification activities that include wind tunnel testing, material strength testing, and deployment tests, supported by CFD analysis and drop tests, as witnessed by NASA in the Mars 2020 Perseverance and Curiosity missions. High-altitude tests, system integration, and flight tests validate this concept.

Verification of the airbag system includes static load testing, inflation timing verification, and deployment testing, supported by drop tests and simulations, as conducted for Mars Pathfinder and ExoMars. Validation includes impact testing, real-life drop testing, and integration testing with the descent/landing system, as was done for the Mars Pathfinder and ExoMars missions. The retro rocket system is verified through thrust vector control, pressure, and thermal stress

tests for accuracy, thrust, and heat resistance. It's validated through unmanned flight tests, integration of systems, and landing simulations as done in NASA's Sky crane, Apollo Lunar Module, and Mars Science Laboratory.

6.1 verification

6.1.1 Static Load Testing

Static load testing is performed to validate that the airbag structure, mounting frames, and load distribution components are designed to support the maximum predicted mechanical forces at touch down with no structural failure. The engineers, by applying controlled and sustained force to the airbag and support assemblies, ensure the materials and weldments meet the required strength margins and do not experience excessive deformation, tearing, or instability under the expected touchdown loads. This test directly supports compliance with the environmental and structural durability requirements, in that it will demonstrate the system can resist worst-case impact pressures without losing structural integrity throughout the mission profile (Daniel Z. Huang, 2019).

6.1.2 Inflation Timing Verification

Inflation timing verification is performed to verify responsiveness and functionality of the Inflation Subsystem to achieve complete deployment within the required time-usually within two seconds of activation and before attainment of readiness altitude. This test will verify valve actuation speed, gas-flow rates, manifold efficiency, and overall inflation dynamics under representative environmental and operational conditions (Emmanuel Siéfert, 2020). The successful verification of this will provide assurance that the timing requirements established by descent safety analyses are met, thereby reducing the risk of partial or delayed deployment, confirming that the airbag system indeed is fully functional prior to ground contact in conformance with reliability and human safety standards set forth.

6.1.3 Deployment Testing

Deployment testing addresses the entire, integrated, end-to-end performance of the airbag system, from firing to inflation, to deployment of structure and stabilization. These tests, in controlled or simulated landings, ensure the deployment of airbags symmetrically, to the correct shape, interacting properly with surrounding structures without entanglement or obstruction. Since the process highlights integration issues, provides redundancy pathways, and ensures that sensors, controllers, and mechanical subsystems work together, deployment testing is required to confirm system-level reliability (Michaela N. Villarreal, 2022). This will help prove that the system is ready to function operationally and will provide proof of compliance to mission safety and verification standards.

6.2 Validation

6.2.1 Full-Scale Landing Drop Test

This test checks that airbags are working as intended in a real-life touchdown, with a drop test of a full-scale airbag performed with a speed that would be needed for a touchdown (Michaela N. Villarreal, 2022). The test also checks that airbags are inflated, absorb impact, re-bounce, and can maintain their inflated pressure for a period of two hours after touchdown.

6.2.2 Integrated Descent Sequence Validation

This test evaluates the overall functional performance of the Landing Control System, consisting of the parachute, airbags, and retrorockets, in a realistic descent test. This test evaluates automatic inflation timing of 1500 m, parachute deceleration, retrorockets for stabilization, and redundancy(Raymond G. Kvaternik, 2013). This test ensures that all subsystems are functioning in a coordinated manner for a successful descent and, hence, a successful landing operation.

6.2.3 Human-in-the- Loop Operational Validation

This test evaluates how safely crew members can operate LCS manual control, warnings, and interfaces under a realistic workload and a spacesuit restriction. This test also verifies the validity of reach, status indication, emergency override, and reaction time of LCS in simulation for a descent operation. The objective of this test is to verify that NF requirements are utilized effectively for descent operation(Trevor Morley and Frank Budnik, 2006).

6.3 Maintenance tasks

The Parachute System undergoes regular analysis to address preventive maintenance with all damages that are identified subjected to repairs or replacement(Stewart, 2013). Adaptive maintenance upgrades the materials and software in line with mission information. Predictive maintenance identifies, in advance, the needs for repairs. The Airbag System is analysed for material integrity and timing of inflation, followed by follow-up actions like component replacement. The adaptive maintenance further optimizes design and software. Predictive activities analyse performance and project potential failures. The Retro Rocket System has routine inspections, which include correction actions such as rocket overhauls and heat shield replacements. Adaptive maintenance improves accuracy, while predictive maintenance follows performance, forecasting when maintenance will be required.

System	Preventive Maintenance	Corrective Maintenance	Adaptive Maintenance	Predictive Maintenance
Parachute System	To inspect materials for wear, tears, or fatigue.	Replace or repair parachutes if damage is found.	Upgrade parachutes with durable materials based on past missions.	To Use test data to predict parachute replacement or repair needs.

Airbag System	To Inspect airbag materials for leaks or degradation.	Replace airbags if punctured or damaged.	Redesign airbags or inflation system based on mission feedback.	To Use sensors to predict failures from landing forces and inflation time.
Retro Rocket	To Inspect components Like nozzles, thrusters, and valves.	Overhaul or replace retro rockets if performance drops.	Integrate new technologies to improve thrust and efficiency.	To Monitor thrust, engine health, and temperature data for maintenance needs.

6.4 Quality requirements

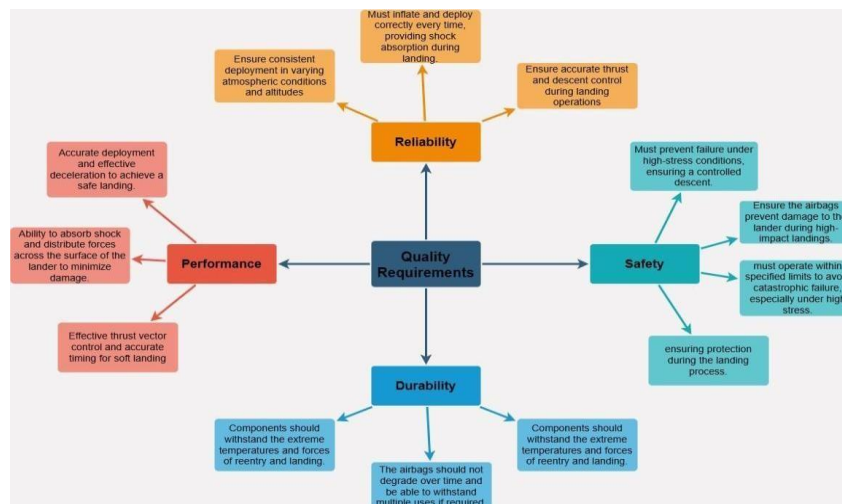


Figure 5: Quality requirements diagram

- **Parachute system:** Tested for structural strength, deployment mechanism, and operation under various environmental conditions. The qualification ensures reliable deployment and compatibility with the descent system at specified altitudes and speeds.

- **Airbag System:** material integrity and shock attenuation tests. Acceptance criteria involve with proper inflation, adequate time for deployment, protection capability during high impact landings.
- **Retro Rocket System:** Qualified by thrust control, pressure, and thermal tests. Ensuring adequate thrust, good control of the descent, and integration with parachutes and avionics, which guarantee a safe landing.

This section covers the qualification and acceptance criteria of the landing system. The criteria outlined are such that each component meets the requirements for safety, performance, durability and reliability in applications where their functioning is critical to the success of a mission.

V-Model

The V-Model in systems development ensures verification and validation throughout the life cycle. Verification activities, which includes system requirements, integration testing, and acceptance testing, make sure that system elements meet the design requirements. Validation activities, including integration and flight testing, confirm the performance of the system in real mission conditions. In NASA's Mars Science Laboratory mission, the V-Model was used to verify parachutes and retro rockets, with successful flight-testing validating system performance during landing.

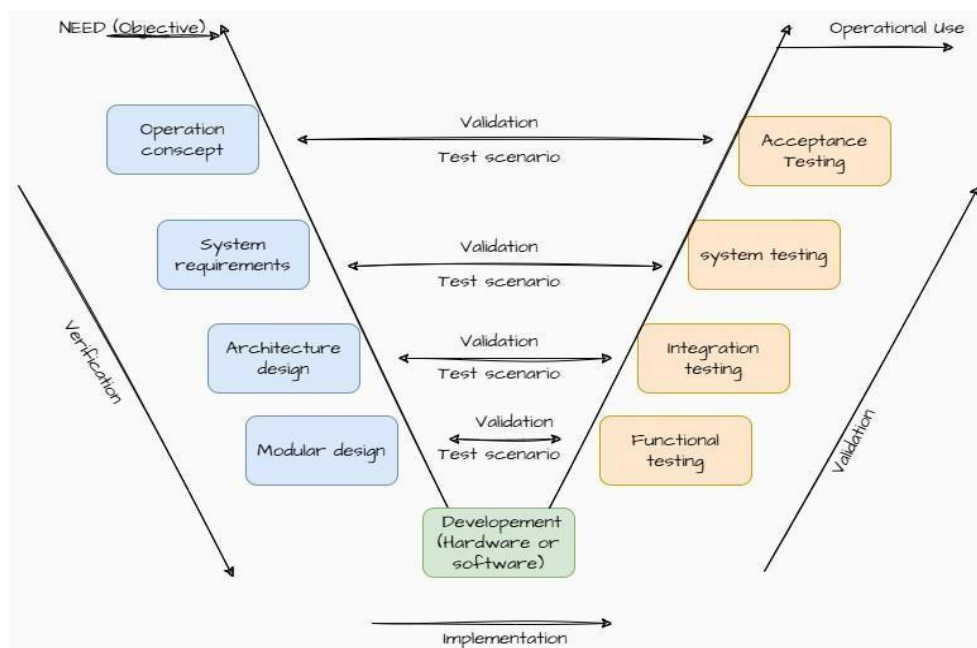


Figure 6: V-Model diagram

Norms and Standards

NASA-STD-5001 – Structural Design and Test Requirements for Space Systems: Ensures structural integrity and reliability for space systems under extreme conditions. Applies directly to retro rockets and parachute systems, ensuring they can withstand the stress during landing(NASA-STD-5001, 1996).

AS9100 – Quality Management System Standard – Aerospace: Establishes requirements for quality management to ensure consistency and high standards throughout the product life of an aerospace product(AS9100D:2016, 2016).

MIL-STD-810 - Environmental Testing: To provide test methods for environmental performance under extreme conditions such as temperature, humidity, vibration, and shock(MIL-STD-810, 1962).

DO-178C - Software Considerations in Airborne Systems: To ensure that safety-critical software is built to rigorous standards for aerospace applications(DO-178C, 2010).

7. Human design

The human–system interface for the Landing Cushion System (LCS) is designed so that astronauts can survive during the landing. The following sections discuss how the design addresses Physical Human Factors, Cognitive Human Factors, and Human Reliability, with concrete examples linked to the existing LCS requirements and architecture.

7.1 Physical Human Factors

a) Seating and restraint configuration matched to touchdown loads

The LCS is designed to keep vertical touchdown velocity at or below 1.524 m/s (5 ft/s), ensuring that peak decelerations remain within spinal and pelvic injury thresholds. To transfer loads through the torso and pelvis rather than the neck or extremities, the crew seats feature 5point restraints and energy-absorbing structures. If the airbags perform poorly or the surface is uneven, this arrangement lowers the risk of compression fractures and soft-tissue injuries. The primary deceleration vector is aligned with the spinal axis by the seat pan angle and backrest recline, which minimizes bending loads. This establishes a direct connection between FR-01 on touchdown velocity and the requirement for impact profiles that are survivable.

b) Reach envelopes and control placement for suited astronauts

Manual LCS controls (e.g. “ARM/INFLATE” and “EMERGENCY VENT”) are located within a 0.6 m reach envelope and within a 30° forward field of view(Dustin M. Gohmert). The panel geometry is based on suited anthropometric data. takes into, reduced shoulder mobility and glove bulk. Large, spaced switch guards and oversized paddles allow operation with pressurised gloves without requiring precise finger dexterity. This layout ensures that, even under high g-loads during final descent, an astronaut strapped into their seat can reach and overrides without leaning, twisting, or unbuckling, which would otherwise increase injury risk.

c) Control forces and posture under impact

The forces needed to operate key controls are chosen so that they are high enough to prevent accidental activation under vibration but low enough for a fatigued astronaut in a pressurised suit to actuate with one hand. The control panel is mounted at chest level to avoid reaching above the shoulder during descent, reducing the risk of shoulder strain. Footrests and sidebracing surfaces allow the crew to maintain a stable posture, preventing injuries and ensuring that when they reach for an override. This alignment of posture, restraint and control forces directly contributes to keeping impact injuries below accepted thresholds.

7.2 Cognitive Human Factors

a) Automation-first concept to manage workload

The LCS mode is fully automatic inflation triggered by onboard sensors, with manual backup. During nominal landing, the crew monitors a compact set of status indicators (“LCS ARMED”, “AUTO INFLATE ACTIVE”, “BAG PRESSURE WITHIN LIMITS”) instead of manually activate valves and gas supplies. This lowers cognitive workload by shifting the crew role from “primary controller” to “supervisory monitor”, allowing them to focus on overall vehicle status, procedures, and contingency preparation rather than fine-grained system management during the most time-critical phase(Dustin M. Gohmert) .

b) Clear, minimal status indications and alarm design

The interface provides visual and auditory cues for the critical LCS states: Ready, Fault, and Deployed. Rather than numerous detailed messages, the design groups underlying sensor and subsystem conditions into a small number of prioritised alerts.

For example:

- A **single high-priority “LCS FAULT – AUTO NOT AVAILABLE”** alarm replaces multiple low-level sensor warnings to avoid alarm flooding.
- recognizable tones and icon shapes differentiate “monitor only” cautions from “action required” warnings.
- In the final 30–60 seconds before touchdown, non-critical alerts are suppressed or logged silently so that only time-critical LCS messages compete for attention.

This alarm strategy directly tackles alarm fatigue, ensuring that when an LCS alert appears, it is clearly associated with a required action.

c) Consistency with broader avionics and crew procedures

The LCS interface uses the same colours, symbology and interaction patterns as the rest of the lander’s avionics displays (for example, consistent icons for “AUTO”, “MANUAL”, and “FAULT”). This reduces the need for specific mental re-mapping during high stress and allowing astronauts to transfer decision making patterns from other subsystems to the LCS.

Crew procedures and checklists mirror the display exactly, so that will help astronaut reads an alert. it is reducing cognitive load.

7.3 Human Reliability

Human reliability in the landing system is enhanced by designing the LCS so that safe performance does not depend on perfect crew actions. it reduces the consequences of human error during high-stress landing operations. Automation manages all functions, with redundant inflation paths and protective logic ensuring that a missed checklist step or incorrect timing does not compromise cushioning performance. features such as guarded switches, physical interlocks, and blocked hazardous command helps prevent mode errors when astronauts are operating under vibration and cognitive load. Manual actions are limited to a small number of clear, high-level commands, supported by two-person verification procedures to minimize accidental activation(Nancy G. Leveson, 1993). Because the interface mirrors the layout and behaviour of the wider avionics system, astronauts benefit from training simulations can accurately replicate real fault scenarios, improving recognition, decision speed, and recovery accuracy. These measures ensure the system anticipates human limitations and mitigates them through simplified workflows, and robust automation.

Conclusion

The Landing Control System (LCS) developed in this project presents a fully integrated and ready solution that ensures a safe and controlled touchdown for a crewed capsule by combining airbags, and retro-rockets into a fault tolerant architecture. Through accurate requirements definition, system modelling, trade-off analysis, and fault tree assessment, the design was shown simplified alternatives and critical failure paths through redundancy, sensor diversity, and fail safe controls. Verification activities including static load tests, inflation timing checks, deployment testing and environmental qualification, giving in to a situation with performance and safety standards. The human system engineering ensured that physical, cognitive and reliability factors rooted into the interface and operational procedures. Overall, the LCS design is validated as feasible, reliable and capable of protecting crew during landing and fully aligning with mission objectives and aerospace engineering requirements.

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